EE 421/521

Project #1 – simple 4-bit calculator

Description : Build a simple calculator which performs addition, subtraction, multiplication, and division of 4-bit signed numbers and demonstrate it on the DE2-115 board. In order to make division operations easy, the fractional part from the division operation will be discarded and only the quotient will be displayed.

Requirements:

1. Sw(7:4) shall be used to determine 4-bit signed value for operand A
2. Sw(3:0) shall be used to determine 4-bit signed value for operand B
3. Operand A shall be displayed on two 7-segment displays (hex(7:6)). For negative numbers, display "-" sign on the display(7). For positive numbers, don't display anything on the display(7).
4. Operand B shall be displayed on two 7-segment displays (hex(5:4)). For negative numbers, display "-" sign on the display(5). For positive numbers, don't display anything on the display(5).
5. Desired operation shall be determined by sw(17:16). "00" = addition, "01" = subtraction, "10" = multiplication, "11" = division.
6. Red LED(17:16) shall indicate the selected operation. "00" = off, off, "01" = off on, "10" = on off, "11" = on on.
7. The selected operation shall be performed when a user pushes button(0).
8. The result from the selected operation shall be displayed on 7-segment displays (2:0). For negative numbers, display "-" sign on the display(2). For positive numbers, don't display anything on the display(2).
9. The result shall be reset to zero when a user pushes button(3) and 7-segment display(2:0) shall display "00".
10. The circuit shall be able to run at minimum 100MHz (check with the timing analyzer tool)
11. A divider module will be supplied. The module shall be integrated with a student's design
12. Double-dabble algorithm converts hex number to Binary Coded Decimal (BCD) number and this algorithm would be provided. The algorithm shall be integrated with a student's design

Grading criteria:

1. Meeting all requirements : 50%
    a. Displaying Inputs on 7-segment displays = 10%
    b. Addition = 5%
    c. Subtraction = 5%
    d. Multiplication = 5%
    e. Division = 5%
    f. Double-dabble algorithm = 10%

    g. Displaying output on 7-segment displays = 10%

2. Design verification : 30%
    a. Testbench setup = 10%
    b. Test cases = 20%

3. Project Report : 20%
    a. Brief description of a design, a top level block diagram, I/O definition = 10%
    b. Others = 10%

Submission:

Archive a whole project directory including constraints files, RTL files, project files (qpf, qsf), simulation script files, and testbench files. Place your report in the “doc” folder.

Due:

10/13/2024 11:59 PM